Software Requirements Specification

Multi-Vendor E-Commerce Marketplace System

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Table of Contents

Section	Title	Page
1	Introduction	3
1.1	Purpose	3
1.2	Scope	3
1.3	Definitions, Acronyms, and Abbreviations	3
1.4	References	3
1.5	Overview	3
2	Overall Description	4
2.1	Product Perspective	4
2.2	Product Functions	4
2.3	User Classes and Characteristics	4
2.4	Operating Environment	4
2.5	Design and Implementation Constraints	4
2.6	User Documentation	4
2.7	Assumptions and Dependencies	4
3	Specific Requirements	5
3.1	External Interfaces	5
3.2	Functional Requirements (FR1–FR12)	5
3.3	Performance Requirements	10
3.4	Logical Database Requirements	10
3.5	Design Constraints	12
3.6	Software System Attributes	12
3.7	Other Requirements	12
4	Appendices	13
A	Role-Based Permission Matrix	13
B	Database Schema Details	13
C	Validation Rules Summary	14
D	Security Implementation Details	14
E	Integration Specifications	15
F	Business Rules and Constraints	15
G	Future Enhancement Roadmap	16
H	API Summary	16
I	Component and Navigation Structure	17
	Version Control	18

1. Introduction

1.1 Purpose

This SRS defines requirements for the Multi-Vendor E-Commerce Marketplace System, a full-featured online marketplace platform enabling multiple sellers to list products, manage orders, and receive payments while providing buyers with a unified shopping experience. Built on Spring Boot 3.x, React.js, JWT, and JPA, the system supports vendor onboarding, product catalog management, multi-cart checkout, payment gateway integration, and marketplace analytics.

1.2 Scope

This system encompasses the following key functionalities:

- Vendor registration, KYC verification, and onboarding workflow management
- Multi-vendor product catalog with category hierarchy, variants, and rich media
- Inventory management with stock alerts and multi-warehouse support
- Buyer registration, profile management, and personalised product recommendations
- Multi-vendor cart and checkout with address management and delivery slot selection
- Payment gateway integration (Razorpay/Stripe) with split settlements to vendors
- Order lifecycle management: placement, confirmation, packing, dispatch, delivery
- Shipping integration with courier APIs for rate calculation and tracking
- Vendor commission management, payout scheduling, and ledger reconciliation
- Ratings, reviews, and Q&A; system for products and vendors
- Dispute resolution and return/refund workflow management
- Marketplace admin dashboard with analytics, revenue reports, and moderation tools

1.3 Definitions, Acronyms, and Abbreviations

Term	Definition
SRS	Software Requirements Specification
SKU	Stock Keeping Unit
PDP	Product Detail Page
PLP	Product Listing Page
GST	Goods and Services Tax
KYC	Know Your Customer
UPI	Unified Payments Interface
NEFT	National Electronic Funds Transfer
COD	Cash on Delivery
EMI	Equated Monthly Instalment

JWT	JSON Web Token
JPA	Java Persistence API
API	Application Programming Interface
SLA	Service Level Agreement
TDS	Tax Deducted at Source
FSSAI	Food Safety and Standards Authority of India
MRP	Maximum Retail Price
RTO	Return to Origin
POD	Proof of Delivery
EDD	Estimated Delivery Date

1.4 References

- IEEE Std 830-1998
- Payment Card Industry Data Security Standard (PCI DSS) v4.0
- GST Act 2017 — Marketplace Seller Compliance
- FSSAI Regulations for Food Products
- Spring Boot 3.x Documentation — https://spring.io
- JWT RFC 7519
- Razorpay API Docs — https://razorpay.com/docs

1.5 Overview

This SRS is organized into four sections: Introduction, Overall Description, Specific Requirements, and Appendices. The Specific Requirements section contains twelve detailed functional requirements (FR1–FR12), each with ten sub-requirements, performance requirements, and database schema. Appendices A through I provide supplementary details on permissions, database schema, validation rules, security implementation, integrations, business rules, future roadmap, API summary, and React component structure.

2. Overall Description

2.1 Product Perspective

The Multi-Vendor E-Commerce Marketplace is a web-based platform on Spring Boot 3.x (REST API port 8080) with React.js frontend (port 8081). JWT with RBAC serves admin, vendor, buyer, delivery partner, and finance roles. Integration with Razorpay for payments, ShipRocket/Delhivery for shipping, Twilio for OTP/SMS, and AWS S3 for product images and documents. PostgreSQL via JPA stores all marketplace data.

2.2 Product Functions

- Vendor onboarding: KYC, GSTIN verification, bank account, and commission setup
- Product catalog: listing, variant management, pricing, and image upload
- Inventory management: stock levels, warehouse mapping, low-stock alerts
- Buyer shopping: search, browse, wishlist, multi-vendor cart, and checkout
- Payment processing: UPI, cards, EMI, COD with split settlement to vendors
- Order management: end-to-end lifecycle from placement to delivery
- Shipping: rate calculation, label generation, pickup scheduling, and tracking
- Return and refund: buyer return request, vendor acceptance, and refund processing
- Vendor payout: commission calculation, TDS deduction, and NEFT/UPI settlement
- Ratings and reviews: product and vendor ratings with moderation
- Dispute management: buyer-vendor disputes with admin arbitration
- Analytics dashboard: GMV, orders, conversion rate, and vendor performance

2.3 User Classes and Characteristics

Marketplace Admin

Manages the entire marketplace: vendor approvals, category management, commission rates, disputes, and platform-wide analytics. Highest privilege role.

Vendor

Registers on the platform, lists products, manages inventory, processes orders, raises shipping pickups, and views their sales dashboard and payout ledger.

Buyer

Browses products, adds to cart, places orders, tracks shipments, writes reviews, and manages returns and refunds.

Delivery Partner

Receives pickup and delivery assignments, updates delivery status, uploads proof of delivery, and manages assigned shipments.

Finance Manager

Reviews vendor payouts, processes settlements, manages TDS certificates, reconciles payment gateway statements, and generates financial reports.

Customer Support

Handles buyer complaints, order issues, and dispute resolution. Views orders and customer profiles without modifying financial data.

2.4 Operating Environment

Backend: Spring Boot 3.x, JDK 17+, PostgreSQL 14+, Redis (session/cart cache). Frontend: React.js 18.x (port 8081). Integrations: Razorpay (payments), ShipRocket/Delhivery (shipping), Twilio (SMS OTP), AWS S3 (images/docs), Elasticsearch (product search). Cloud: AWS with Auto Scaling Groups.

2.5 Design and Implementation Constraints

- PCI DSS compliance: card data never stored; tokenised via Razorpay vault
- GST compliance: tax calculation per product HSN code; GSTIN validation via government API
- Product images stored on AWS S3; max 5 images per SKU, each 5MB
- Vendor GSTIN and bank account must be verified before first product listing
- Commission rates configured per category; overridable per vendor by admin
- Spring Boot 3.x, Java 17+; React.js functional components; REST API only
- HTTPS-only; JWT auth; CORS restricted to frontend origin
- Maximum cart size: 20 distinct SKUs from 10 vendors per checkout session

2.6 User Documentation

Vendor Onboarding Guide, Buyer Help Center, Admin Operations Manual, Finance Manager Payout Guide, Shipping Integration Guide, API Documentation (Swagger).

2.7 Assumptions and Dependencies

- Razorpay merchant account and API keys provisioned before go-live
- ShipRocket/Delhivery courier partner agreements in place
- Government GSTIN verification API access available
- AWS S3 bucket provisioned for product images and KYC documents
- SMS OTP via Twilio; email via SES for transactional notifications

3. Specific Requirements

3.1 External Interfaces

3.1.1 User Interfaces

Interface	Description	Technology
Buyer Storefront	PLP, PDP, cart, checkout, order tracking	React.js + Redux
Vendor Dashboard	Product management, orders, inventory, payouts	React.js
Admin Console	Approvals, commissions, disputes, analytics	React.js + charts
Finance Portal	Payout processing, reconciliation, TDS	React.js
Delivery App	Assignment list, delivery updates, POD upload	React.js PWA

3.1.2 Hardware Interfaces

Standard cloud servers (8 vCPU, 32GB RAM for app; 16 vCPU, 64GB RAM for DB); AWS S3 for media; Redis cluster for cart and session caching.

3.1.3 Software Interfaces

- Razorpay — payment capture, split settlements, refunds, and webhook events
- ShipRocket / Delhivery API — rate calculation, AWB generation, pickup, and tracking
- Twilio — SMS OTP for buyer/vendor registration and order notifications
- AWS S3 — product image and KYC document storage
- Elasticsearch 8.x — full-text product search with filters and facets
- Government GSTIN API — vendor GSTIN verification at onboarding

3.1.4 Communication Interfaces

REST over HTTPS; JWT Bearer token auth; Webhook listeners for Razorpay payment events and ShipRocket tracking updates; WebSocket for real-time order status updates; JSON data exchange.

3.2 Functional Requirements

FR1: Vendor Registration and Onboarding

The system shall provide a complete vendor onboarding workflow including business registration, KYC document upload, GSTIN verification, bank account setup, and marketplace agreement acceptance.

Detailed Requirements:

- Vendor self-registration: business name, type (individual/proprietorship/company), email, mobile

• KYC document upload: Aadhaar/PAN for individual, GST certificate and incorporation for company

• GSTIN verification: real-time validation via government API with status feedback

• Bank account addition: account number, IFSC, account holder name — verified via penny drop

• Marketplace agreement: digital acceptance of commission terms and seller policy

• Admin approval workflow: KYC review, approval, or rejection with reason

• Vendor profile: store name, logo, banner, description, and return policy

• Commission tier assignment: per-category commission % configured by admin at onboarding

• Vendor dashboard activation: product listing enabled only after admin approval

• Welcome email and onboarding checklist: 5 steps to first product listing

FR2: Product Catalog Management

The system shall enable vendors to create, manage, and publish product listings with full attribute management, variant support, rich media, and pricing.

Detailed Requirements:

• Product creation: title, description, category, sub-category, brand, HSN code, and MRP

• Variant management: size, colour, material — each variant has distinct SKU, price, and stock

• Image upload: up to 5 images per SKU; drag-and-drop reorder; auto-thumbnail generation

• Video support: product demo video URL (YouTube/Vimeo embed) per listing

• Pricing: selling price, MRP, and optional discount percentage; GST rate auto-applied by HSN

• Bulk import: CSV/Excel upload for batch product creation with validation report

• Category hierarchy: admin-managed 3-level category tree (category sub-category type)

• Product status: Draft, Pending Review, Active, Inactive — with transition rules

• Admin review: flagged products (below quality score) queued for manual review

• SEO fields: meta title, meta description, and URL slug per product

FR3: Inventory Management

The system shall provide real-time inventory tracking per SKU with multi-warehouse support, stock alerts, and automated out-of-stock handling.

Detailed Requirements:

• Per-SKU stock quantity tracked across one or more warehouse locations

• Multi-warehouse: vendors with multiple fulfilment centres assign stock per location

• Stock update: manual adjustment or bulk CSV upload; reason code required for adjustments

• Low-stock alert: configurable threshold per SKU; email and dashboard alert when breached

• Out-of-stock handling: listing auto-set to 'Out of Stock' when quantity = 0

• Back-in-stock notification: buyers who wishlisted OOS items notified on restock

• Reserved stock: stock reserved on order placement; released on cancellation

- Inventory report: stock level, movement history, and turnover rate per SKU
- Shrinkage tracking: damaged/lost stock recorded with reason for audit trail
- Inventory sync: optional API integration for vendors with own ERP/WMS systems

FR4: Buyer Experience and Search

The system shall provide buyers with a personalised, high-performance shopping experience including search, browse, wishlist, and product comparison.

Detailed Requirements:

- Buyer registration: email/mobile OTP registration with profile creation
- Full-text product search via Elasticsearch: keyword search with spell correction
- Search filters: price range, brand, rating, delivery time, vendor, discount %
- Faceted navigation on PLP: filter by attribute values (size, colour, material)
- Personalised recommendations: ML-based 'Customers also bought' and 'Recommended for you'
- Wishlist: save products for later; share wishlist via link
- Product comparison: compare up to 4 products side-by-side on attribute table
- Recently viewed: last 20 viewed products displayed on homepage and PDP
- Address book: multiple saved delivery addresses with default selection
- Guest checkout: buy without registration with order tracked via email/mobile

FR5: Cart and Checkout

The system shall support a multi-vendor cart, address selection, delivery slot booking, coupon application, and secure checkout flow.

Detailed Requirements:

- Multi-vendor cart: products from multiple vendors in single cart; grouped by vendor at checkout
- Quantity management: increase/decrease or remove items; real-time price recalculation
- Address selection: pick from saved addresses or add new; map-based address validation
- Delivery slot: available delivery windows per vendor/pincode with slot booking
- Coupon/promo code: apply codes with validation; cashback, flat discount, % off types
- Order summary: itemised price breakdown — item total, discount, shipping, GST, grand total
- Payment method: UPI, debit/credit card, net banking, EMI, wallet, COD per pincode
- Razorpay checkout: tokenised card storage; 3D Secure authentication
- Split order: single checkout creates separate sub-orders per vendor for fulfilment
- Order confirmation: email and SMS with order ID, estimated delivery date, and summary

FR6: Order Management

The system shall manage the complete order lifecycle from placement through packing, dispatch, delivery, and post-delivery resolution.

Detailed Requirements:

- Order states: Placed Confirmed Packed Dispatched Out for Delivery Delivered / Cancelled / Returned
- Vendor order dashboard: incoming orders with accept/reject (within SLA) action
- Packing slip: auto-generated PDF with order items, buyer address, and barcode
- Dispatch: vendor enters AWB number or triggers ShipRocket pickup; tracking activated
- Buyer order tracking: real-time map-based tracking with courier partner updates
- Order cancellation: buyer can cancel before dispatch; vendor can cancel with reason
- SLA tracking: vendor response SLA (30 min) and dispatch SLA (24/48 hr) with breach alerts
- Partial order: if one vendor's items cancelled, remaining sub-orders unaffected
- Delivery confirmation: courier updates on delivery; buyer OTP confirmation optional
- Order history: complete list with filters (status, date, vendor) exportable as CSV

FR7: Payment Processing and Settlement

The system shall handle multi-vendor payment capture, marketplace commission deduction, TDS computation, and vendor payout settlement.

Detailed Requirements:

- Payment capture: Razorpay route captures full buyer payment into marketplace escrow
- Split settlement: Razorpay route or manual split — vendor amount = order value commission TDS
- Commission calculation: per-category % applied at order level; configurable per vendor
- TDS deduction: 1% TDS on seller payouts per Income Tax Section 194-O
- Payout schedule: weekly payout cycle (configurable); settlement after 7-day return window
- Payout ledger: itemised credit/debit entries per order, return, and adjustment
- NEFT/UPI payout: bulk payout via bank transfer; confirmation with UTR reference
- COD reconciliation: vendor remits COD collected; system reconciles against expected amount
- Payout report: downloadable Excel/CSV with all transactions for vendor accounting
- TDS certificate: Form 16A generated quarterly per vendor for tax filing

FR8: Shipping Management

The system shall integrate with courier partners for rate calculation, label generation, pickup scheduling, real-time tracking, and delivery exception handling.

Detailed Requirements:

- Shipping rate calculation: multi-courier rate comparison by weight, dimensions, and pincode
- Courier selection: vendor selects preferred courier or system auto-assigns on lowest rate
- AWB generation: electronic airway bill created via ShipRocket API on dispatch

• Label printing: shipping label PDF generated with barcode, addresses, and order details

• Pickup scheduling: vendor schedules pickup slot; courier confirms pickup

• Real-time tracking: tracking events (picked up, in transit, out for delivery) from courier webhook

• RTO handling: undelivered shipments returned to vendor; RTO status updated in order

• NDR management: non-delivery report with reason; re-attempt or return triggered

• Delivery SLA: platform-level EDD shown to buyer based on vendor warehouse to buyer pincode

• POD upload: delivery partner uploads signature or OTP-based proof of delivery

FR9: Ratings, Reviews, and Q&A;

The system shall enable buyers to rate products and vendors, write reviews, upload photos, and ask product questions answered by vendors or other buyers.

Detailed Requirements:

• Product rating: 1–5 star rating post-delivery with optional review title and text

• Photo/video review: buyers upload up to 3 photos or 1 video with review

• Verified purchase badge: review tagged as 'Verified Purchase' if from confirmed buyer

• Vendor rating: separate vendor rating for shipping speed, packaging, and communication

• Review moderation: ML-based spam detection; admin moderation queue for flagged reviews

• Helpful votes: buyers vote 'Helpful' on reviews; top reviews shown prominently

• Vendor response: vendor can publicly reply to buyer reviews

• Q&A; section: buyers post product questions; vendor or community answers

• Rating summary: average rating, rating distribution bar chart on PDP

• Review report: vendors view ratings analytics — average, trend, and response rate

FR10: Return and Refund Management

The system shall manage the end-to-end return and refund process from buyer return request through vendor inspection, courier pickup, and refund processing.

Detailed Requirements:

• Return request: buyer initiates within return window (7/15/30 days per category)

• Return reason: dropdown (wrong item, damaged, not as described, changed mind) with photo upload

• Vendor review: vendor accepts or rejects return with reason; admin arbitrates disputes

• Return pickup: courier scheduled for reverse pickup from buyer address

• Item inspection: vendor marks item as received and condition (acceptable / damaged)

• Refund initiation: original payment method refund triggered after inspection approval

• Refund modes: source account refund (3–5 days) or platform wallet (instant)

• Vendor debit: return amount debited from vendor payout ledger; commission partially reversed

- Return report: vendor views return rate by product with root cause analysis
- Platform return SLA: end-to-end return resolved within 10 business days

FR11: Dispute Resolution

The system shall provide a structured dispute management system for buyer-vendor conflicts with escalation to marketplace admin and binding resolution.

Detailed Requirements:

- Dispute trigger: buyer raises dispute after vendor rejects return or fails to respond
- Dispute types: item not delivered, item not as described, refund not received, vendor unresponsive
- Evidence submission: both parties upload photos, chat screenshots, and documents
- Dispute timeline: vendor response within 48 hours; admin intervention if no response
- Admin arbitration: admin reviews evidence and issues binding resolution
- Resolution options: full refund, partial refund, replacement, or dispute rejected
- Vendor penalty: repeated disputes trigger escalating penalties (warning fee suspension)
- Dispute status tracking: buyer and vendor see current status and next action
- Communication log: all messages between buyer, vendor, and admin logged per dispute
- Dispute analytics: admin views dispute rate by vendor and category for quality monitoring

FR12: Marketplace Analytics and Reporting

The system shall provide role-specific analytics dashboards with GMV trends, conversion funnels, vendor performance, and financial summaries.

Detailed Requirements:

- Admin GMV dashboard: total gross merchandise value, orders, AOV, and net revenue trend
- Conversion funnel: sessions PDP views add-to-cart checkout orders (daily/weekly)
- Vendor performance scorecard: GMV, orders, return rate, rating, and SLA breach count
- Category analytics: top categories by GMV, units, and margin contribution
- Vendor sales report: downloadable order-level report with commission and payout breakdown
- Buyer analytics: new vs returning buyers, LTV, and cohort retention charts
- Search analytics: top search terms, zero-result queries, and click-through rate
- Inventory health: slow-moving SKUs, OOS frequency, and days of stock remaining
- Payment analytics: payment method mix, failure rate, and refund rate
- Custom date range: all dashboards filterable by date range with comparison to prior period

3.3 Performance Requirements

Metric	Requirement	Measurement Method
PLP Load Time	< 1.5 seconds (50 products)	Load test

Search Response	< 500ms for 95th percentile	Elasticsearch benchmark
Checkout API	< 800ms end-to-end	Performance test
Payment Capture	< 3 seconds via Razorpay	Integration test
Concurrent Users	50,000 simultaneous shoppers	Stress test
Image CDN Load	< 300ms per image	CDN benchmark
Order Placement TPS	500 orders/second at peak	Load test
System Uptime	99.95% availability	Uptime monitoring
Search Index Freshness	New products indexed < 5 min	Search test
Payout Processing	10,000 payouts processed/hour	Batch test

3.4 Logical Database Requirements

vendors

Description: Vendor/seller profiles

Fields: vendor_id (PK UUID), business_name, gstin (VARCHAR 15 UNIQUE), pan (VARCHAR 10), bank_account_id (FK), status (ENUM: PENDING/APPROVED/SUSPENDED), commission_tier_id (FK), store_name, store_slug (UNIQUE), kyc_approved_at, created_at, updated_at

products

Description: Product master catalog

Fields: product_id (PK UUID), vendor_id (FK), title (VARCHAR 300), description (TEXT), category_id (FK), brand, hsn_code (VARCHAR 10), mrp (DECIMAL 10,2), selling_price (DECIMAL 10,2), gst_rate (DECIMAL 5,2), status (ENUM: DRAFT/PENDING/ACTIVE/INACTIVE), avg_rating (DECIMAL 3,2), review_count (INTEGER), created_at, updated_at

product_variants

Description: SKU variants of products

Fields: variant_id (PK UUID), product_id (FK), sku (VARCHAR 50 UNIQUE), size (VARCHAR 50), colour (VARCHAR 50), material (VARCHAR 50), price (DECIMAL 10,2), stock_quantity (INTEGER DEFAULT 0), reserved_qty (INTEGER DEFAULT 0), images (JSONB), is_active (BOOLEAN)

orders

Description: Master order per checkout

Fields: order_id (PK UUID), buyer_id (FK), total_amount (DECIMAL 12,2), payment_status (ENUM: PENDING/PAID/FAILED/REFUNDED), payment_method, razorpay_payment_id

(VARCHAR 100), coupon_id (FK nullable), shipping_address_id (FK), created_at, updated_at

sub_orders

Description: Per-vendor order split

Fields: sub_order_id (PK UUID), order_id (FK), vendor_id (FK), status (ENUM: PLACED/CONFIRMED/PACKED/DISPATCHED/DELIVERED/CANCELLED/RETURNED), awb_number (VARCHAR 50 nullable), courier_id (FK nullable), dispatch_at (TIMESTAMP), delivered_at (TIMESTAMP), commission_amount (DECIMAL 10,2), payout_amount (DECIMAL 10,2)

cart_items

Description: Buyer cart items (Redis-backed)

Fields: cart_item_id (PK UUID), buyer_id (FK), variant_id (FK), quantity (INTEGER), price_at_add (DECIMAL 10,2), added_at (TIMESTAMP)

reviews

Description: Product reviews

Fields: review_id (PK UUID), product_id (FK), buyer_id (FK), sub_order_id (FK), rating (INTEGER 1-5), title (VARCHAR 200), body (TEXT), photo_urls (JSONB), is_verified_purchase (BOOLEAN), helpful_count (INTEGER), status (ENUM: PENDING/APPROVED/REJECTED), created_at

returns

Description: Return requests

Fields: return_id (PK UUID), sub_order_id (FK), buyer_id (FK), reason (ENUM: WRONG_ITEM/DAMAGED/NOT_AS_DESCRIBED/CHANGED_MIND/OTHER), status (ENUM: REQUESTED/VENDOR_APPROVED/PICKUP_SCHEDULED/RECEIVED/REFUNDED/REJECTED), refund_amount (DECIMAL 10,2 nullable), reverse_awb (VARCHAR 50 nullable), created_at

disputes

Description: Buyer-vendor disputes

Fields: dispute_id (PK UUID), return_id (FK nullable), sub_order_id (FK), buyer_id (FK), vendor_id (FK), type (ENUM: NOT_DELIVERED/NOT_AS_DESCRIBED/REFUND_NOT_RECEIVED/UNRESPONSIVE), status (ENUM: OPEN/VENDOR_RESPONDED/ADMIN_REVIEW/RESOLVED/CLOSED), resolution (ENUM nullable), resolved_by (FK users nullable), resolved_at (TIMESTAMP)

payouts

Description: Vendor payout records

Fields: payout_id (PK UUID), vendor_id (FK), cycle_start (DATE), cycle_end (DATE), gross_amount (DECIMAL 12,2), commission_deducted (DECIMAL 10,2), tds_deducted (DECIMAL 10,2), net_amount (DECIMAL 12,2), status (ENUM: PENDING/PROCESSING/PAID/FAILED), utr_reference (VARCHAR 50 nullable), paid_at (TIMESTAMP)

categories

Description: Product category hierarchy

Fields: category_id (PK UUID), name, parent_id (FK nullable), level (INTEGER 1-3), commission_default (DECIMAL 5,2), gst_default (DECIMAL 5,2), is_active (BOOLEAN)

coupons

Description: Coupon/promo codes

Fields: coupon_id (PK UUID), code (VARCHAR 30 UNIQUE), type (ENUM: FLAT/PERCENT/CASHBACK), value (DECIMAL 10,2), min_order_value (DECIMAL 10,2), max_discount (DECIMAL 10,2 nullable), valid_from (DATE), valid_to (DATE), usage_limit (INTEGER), used_count (INTEGER DEFAULT 0), is_active (BOOLEAN)

3.5 Design Constraints

- Card data never stored; all card transactions tokenised via Razorpay Vault
- GST calculated server-side based on HSN code; never trusted from frontend
- Vendor GSTIN and bank account verified before any payout is processed
- Product images served via CloudFront CDN; direct S3 URLs not exposed to browser
- Stock reservation transactional: concurrent checkout must not oversell
- Spring Boot 3.x, Java 17+; React.js functional components; Elasticsearch for search
- HTTPS-only; JWT auth; CORS restricted; CSP headers enforced
- All financial transactions immutable in ledger; adjustments via contra entries only

3.6 Software System Attributes

3.6.1 Reliability

99.95% uptime; ACID transactions for stock reservation and payment; hourly DB snapshots; CDN failover for media.

3.6.2 Availability

24×7 availability; maintenance windows 02:00–04:00 IST Sunday; Auto Scaling for peak sale events.

3.6.3 Security

PCI DSS compliance; AES-256 at rest; TLS 1.2+; JWT auth; RBAC; Razorpay Vault for cards; VAPT bi-annually.

3.6.4 Maintainability

Clean architecture; JUnit 5 80% coverage; Flyway migrations; Swagger API docs; modular React with lazy loading.

3.6.5 Portability

Docker + Kubernetes on AWS EKS; Elasticsearch managed service; S3-compatible storage abstraction.

3.7 Other Requirements

Compliance: PCI DSS for payments; DPDP Act 2023 for buyer data; GST Act for tax invoices. Accessibility: WCAG 2.1 AA for storefront. Languages: English + regional (configurable). Tax invoice generated per sub-order for GST compliance.

4. Appendices

Appendix A: Role-Based Permission Matrix

Feature	Admin	Vendor	Buyer	Finance Mgr	Support	Delivery
Approve Vendors						

- order_id: UUID NOT NULL REFERENCES orders(order_id)
- vendor_id: UUID NOT NULL REFERENCES vendors(vendor_id)
- status: ENUM('PLACED','CONFIRMED','PACKED','DISPATCHED','DELIVERED','CANCELLED','RETURNED')
- awb_number: VARCHAR(50)
- courier_id: UUID REFERENCES couriers
- dispatch_at: TIMESTAMP
- delivered_at: TIMESTAMP
- commission_rate: DECIMAL(5,2) NOT NULL
- commission_amount: DECIMAL(10,2) NOT NULL
- tds_rate: DECIMAL(5,2) DEFAULT 1.0
- tds_amount: DECIMAL(10,2)
- payout_amount: DECIMAL(10,2)
- payout_id: UUID REFERENCES payouts nullable

B.3 payouts Table

- payout_id: UUID PRIMARY KEY
- vendor_id: UUID NOT NULL REFERENCES vendors(vendor_id)
- cycle_start: DATE NOT NULL
- cycle_end: DATE NOT NULL
- gross_amount: DECIMAL(12,2) NOT NULL
- commission_deducted: DECIMAL(10,2) NOT NULL
- tds_deducted: DECIMAL(10,2) NOT NULL
- net_amount: DECIMAL(12,2) NOT NULL
- status: ENUM('PENDING','PROCESSING','PAID','FAILED') DEFAULT 'PENDING'
- bank_account_id: UUID NOT NULL
- utr_reference: VARCHAR(50)
- paid_at: TIMESTAMP

Appendix C: Validation Rules Summary

Field/Rule	Validation Criterion	Error Message
GSTIN	15-char alphanumeric per govt format	Invalid GSTIN format
Product MRP	Must be > 0 and selling price	MRP cannot be less than selling price
Stock Quantity	Non-negative integer	Stock cannot be negative
Coupon Code	Uppercase alphanumeric 4–12 chars	Invalid coupon format
Return Window	Request within 7/15/30 days of delivery	Return window expired
Order Amount	I 1 (min order value)	Order amount too low
Commission Rate	0–50% per category	Commission rate out of range

Image Size	Each image 5MB; max 5 per SKU	Image too large or too many
Pincode	6-digit Indian pincode	Invalid pincode
Review Rating	Integer 1–5 only	Rating must be between 1 and 5

Appendix D: Security Implementation Details

Parameter	Value
Algorithm	HS512
Access Token Expiry	3600s (buyer) / 7200s (vendor)
Refresh Token Expiry	604800s (7 days)
Card Data Storage	Razorpay Vault only — never in system DB
Image Storage	AWS S3 with CloudFront CDN
PCI DSS	Compliant — tokenised payments only

Exception Handling Classes

- GlobalExceptionHandler — RFC 7807 problem+json
- OutOfStockException — stock reserved exceeds available (HTTP 409)
- VendorNotApprovedException — listing attempt before approval (HTTP 403)
- InvalidGSTINException — GSTIN format or API validation failure (HTTP 400)
- PaymentCaptureException — Razorpay payment failure (HTTP 402)
- ReturnWindowExpiredException — return request after deadline (HTTP 400)
- CouponExpiredException — coupon past validity or limit reached (HTTP 400)
- DuplicateSKUException — duplicate SKU code on product creation (HTTP 409)

Appendix E: Integration Specifications

E.1 Razorpay Payment Gateway

Full payment lifecycle: order creation, checkout, capture, split route to marketplace account, refunds, and webhook events (payment.captured, payment.failed, refund.processed). COD orders marked pending; payment captured on delivery confirmation.

E.2 ShipRocket Courier API

Rate calculation (multi-courier), AWB generation, pickup scheduling, label download, real-time tracking webhooks (shipment_track_update, delivery_done, RTO_initiated). Supports Delhivery, BlueDart, DTDC, and 20+ couriers.

E.3 Government GSTIN Verification

REST API call to government portal on vendor GSTIN entry; validates GSTIN existence, business name match, and registration status. Caches result for 24 hours.

Appendix F: Business Rules and Constraints

F.1 Inventory and Pricing Rules

- Selling price must be MRP at all times; system rejects listings violating this
- Stock reservation is transactional — concurrent orders cannot oversell the same SKU
- Commission is calculated on selling price excluding GST
- TDS at 1% deducted on gross payout to vendor per Section 194-O of Income Tax Act

F.2 Order and Return Rules

- Vendor must confirm order within 30 minutes; auto-cancellation if no response
- Return window resets to 15 days if replacement is issued (not original window)
- Payout is held for 7 days post-delivery to cover return window before settlement
- Vendor suspension triggered after dispute rate > 5% of orders in 30-day rolling period

Appendix G: Future Enhancement Roadmap

Phase 2 (Q3 2026)

- Hyperlocal delivery: real-time dark store integration for 2-hour delivery in metro cities
- Voice search: vernacular voice search in 8 Indian languages via Google Speech API

Phase 3 (Q1 2027)

- Live commerce: vendor livestream with in-stream product purchase and real-time Q&A;
- ONDC integration: open network for digital commerce participation as buyer-side and seller-side app

Phase 4 (Q3 2027)

- AI personalisation engine: real-time recommendation using collaborative filtering and embeddings
- Buy Now Pay Later: BNPL integration with 3–6 month 0% EMI options via fintech partners

Appendix H: API Summary

H.1 Vendor and Product APIs

Method	Endpoint	Description	Access
POST	/api/v1/vendors/register	Vendor registration	Public
PUT	/api/v1/vendors/{id}/approve	Approve vendor	Admin
POST	/api/v1/products	Create product listing	Vendor
PUT	/api/v1/products/{id}/publish	Publish product	Vendor
GET	/api/v1/products/search	Search products	Public

H.2 Order and Payment APIs

Method	Endpoint	Description	Access
POST	/api/v1/cart/checkout	Initiate checkout	Buyer
POST	/api/v1/orders	Place order	Buyer
POST	/api/v1/payments/capture	Capture payment	System
PUT	/api/v1/sub-orders/{id}/dispatch	Mark dispatched	Vendor
POST	/api/v1/returns	Raise return request	Buyer

H.3 Payout and Analytics APIs

Method	Endpoint	Description	Access
GET	/api/v1/payouts/vendor/{id}	Vendor payout ledger	Vendor/Finance
POST	/api/v1/payouts/process	Trigger payout batch	Finance
GET	/api/v1/analytics/gmv	GMV dashboard	Admin
GET	/api/v1/analytics/vendor/{id}	Vendor scorecard	Admin/Vendor
POST	/api/v1/disputes	Raise dispute	Buyer

Appendix I: Component and Navigation Structure

I.1 App.jsx

Redux store for cart/auth, React Router v6 with role-based route guards, Razorpay script loader, JWT interceptor, real-time order status WebSocket.

I.2 NavBar.jsx

Logo, search bar (Elasticsearch), category menu, wishlist (badge), cart (item count badge), notifications bell, profile dropdown. Vendor/Admin nav variant.

I.3 Home.jsx

Hero banner carousel, featured categories grid, personalised product recommendations, flash deals countdown timer, trending products, and recently viewed section.

I.4 Marketplace.jsx

PLP with infinite scroll + filters panel | PDP with image gallery, variants selector, add-to-cart, ratings summary, review list, Q&A; | Cart page | Checkout wizard (address delivery payment) | Order tracking map | Vendor dashboard (products table, orders kanban, inventory, payouts, analytics) | Admin console (vendor approvals, disputes, category, commissions, platform analytics).

I.5 Footer.jsx

Category links, seller registration CTA, buyer protection policy, GST invoice info, payment logos (Visa, Mastercard, UPI, Razorpay), social media links, copyright.

Document Version Control

Version	Date	Author	Changes
1.0	June 2026	iamneo Team	Initial Release
1.1	TBD	iamneo Team	Review Comments Incorporated
2.0	TBD	iamneo Team	Post-UAT Updates